

Clarkson Institute Summer School: Examples Sheet 8 - Permutation representations and linear representations

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Permutation representations

1. Consider the permutation representation $f : D_4 \rightarrow \text{Sym}(\{1, 2, 3, 4\})$ where $\{1, 2, 3, 4\}$ are the vertices of a square. Identify the orbit and stabiliser of 1, and check that the orbit-stabiliser theorem is valid.

Solution: We have $\text{Orb}(1) = \{1, 2, 3, 4\}$. The stabiliser is $\text{Stab}(1) = \{e, rm\}$, where rm is a reflection in the diagonal passing through 1 and the opposite vertex. Since $4 \cdot 2 = 8$, the orbit stabiliser theorem is valid.

2. We define the *conjugation action* to be the map $f : G \rightarrow \text{Sym}(G)$ given by $f(g) = \alpha_g$, where $\alpha_g(h) = ghg^{-1}$. Show that this indeed gives a permutation representation. Identify the orbit and stabiliser of a given element $x \in G$ with an object from earlier in the course. What does the orbit-stabiliser theorem tell you in this case?

Solution: First, note that $\alpha_{g^{-1}} = \alpha_g^{-1}$ so that α_g is invertible and thus a permutation of G . Next, observe that:

$$\alpha_{gg'}(h) = (gg')h(gg')^{-1} = gg'h(g')^{-1}g = \alpha_g(\alpha_{g'}(h)),$$

which gives $f(gg') = \alpha_{gg'} = \alpha_g \circ \alpha_{g'} = f(g)f(g')$. Hence, f is a homomorphism from G to $\text{Sym}(G)$, and thus a permutation representation as required.

The orbit of x is the set of all elements which are conjugate to x , i.e. $\text{Orb}(x) = \text{Cl}(x)$, i.e. the conjugacy class. The stabiliser of x is the set of all conjugating elements which fix x under conjugation, i.e. the centraliser $\text{Stab}(x) = C(x)$. In particular, the orbit-stabiliser theorem tells us that the size of the conjugacy class and the size of the centraliser divides the group order.

3. Suppose that G is a finite simple group of order greater than two, and suppose that $H \leq G$ is a proper subgroup with n cosets in G . Show that $|G| \leq n!$.

Solution: Consider the left coset action, given by $f : G \rightarrow \text{Sym}(G/H) \cong S_n$ where $f(g) = \alpha_g$ with $\alpha_g(g'H) = (gg')H$. We saw in the lectures that this is a permutation representation. Since G is a simple group, the kernel of this homomorphism must be $\text{Ker}(f) = \{e\}$ or $\text{Ker}(f) = G$.

However, since $H \leq G$ is a proper subgroup, we have $H \neq G$, and thus H has at least two cosets. Further, we have $\text{Orb}(gH) = G/H$, since we can map g_1H to g_2H by $\alpha_{g_2g_1^{-1}}$. Thus $\text{Ker}(f) \neq G$ (not every permutation is the identity), and thus f is an injection and G is isomorphic to a subgroup of $\text{Sym}(G/H) \cong S_n$. It follows that $|G| \leq n!$.

Definitions and examples of linear representations

4. In each of the following cases, decide whether or not the following assignments of matrices to group generators extend to representations. Justify your answers.

(a) $D : \mathbb{Z} \rightarrow GL(2, \mathbb{C})$, given by $D(1) = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$;

(b) $D : \mathbb{Z} \rightarrow GL(2, \mathbb{C})$, given by $D(1) = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$;

(c) $D : \mathbb{Z}_4 \rightarrow GL(2, \mathbb{C})$, given by $D(1) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$;

(d) $D : S_3 \rightarrow GL(2, \mathbb{C})$, given by $D((12)) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$, $D((23)) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$.

Of those that are representations, which are faithful?

Solution:

(a) This extends to a representation. We have:

$$D(n) = D(1)^n = \begin{pmatrix} 2^n & 0 \\ 0 & 3^n \end{pmatrix}.$$

So in particular, we see that $D(n+m) = D(n)D(m)$. This is faithful, since the map is injective.

(b) This extends to a representation. We have:

$$D(2) = D(1)^2 = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}.$$

By an inductive proof, we can show:

$$D(n) = D(1)^n = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix},$$

which matches the two-dimensional representation we saw in the lecture.

(c) We check the relation $D(1)^4 = I$. We have:

$$D(1)^2 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}.$$

This gives $D(1)^4 = I$, as required. Thus this extends to a representation. It is faithful since none of $D(1), D(1)^2, D(1)^3$ or I are equal.

(d) Identify (12) with a reflection rm of a triangle, and (23) as another reflection m . Then their product $(12)(23) = (123)$ should correspond to the rotation r . Therefore if we define:

$$D((123)) = D((12))D((23)) = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix},$$

we just need to check the usual relations $r^3 = e$, $m^2 = e$ and $rm = mr^{-1}$. But observe that $D((123))^3$ is given by:

$$D((123))^3 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} -1 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \neq I.$$

Thus the relations do not hold, and this cannot extend to a representation.

5. Find all one-dimensional representations of: (a) D_n ; (b) Q_8 .

Solution:

- (a) D_n is generated by r, m subject to the relations $r^n = e$, $m^2 = e$ and $rm = mr^{-1}$. Since $\mathbb{C}^*$ is commutative, any representation $D : D_n \rightarrow \mathbb{C}^*$ implies that $D(r)D(m) = D(m)D(r)^{-1}$ can be rewritten as $D(r)^2 = 1$. Thus the relations we need are:

$$D(r)^n = 1, \quad D(m)^2 = 1, \quad D(r)^2 = 1.$$

The second two relations imply that $D(m) = \pm 1$, $D(r) = \pm 1$. The first relation implies that $D(r)^n = (\pm 1)^n = 1$, so we see that if n is even both options work, but if n is odd we require $D(r) = 1$. This gives the two cases:

- If n is odd, we have $D(r) = 1$ and $D(m) = \pm 1$ as the two possible representations.
 - If n is even, we have $D(r) = \pm 1$ and $D(m) = \pm 1$ generating four possible representations, with all possible choices available.
- (b) Q_8 is generated by i, j subject to the relations $i^4 = 1$, $i^2 = j^2$ and $ij = ji^{-1}$. One again, since $\mathbb{C}^*$ is commutative, any representation $D : Q_8 \rightarrow \mathbb{C}^*$ implies that $D(i)D(j) = D(j)D(i)^{-1}$ can be rewritten as $D(i)^2 = 1$. This gives the relations:

$$D(i)^4 = 1, \quad D(i)^2 = D(j)^2 = 1.$$

The first relation is implied by the remaining relations. Thus the possibilities are $D(i) = \pm 1$, $D(j) = \pm 1$, giving four different possible representations.

6. Construct the left regular representations of C_2 and C_3 .

Solution: Let $C_2 = \{e, r\}$. The permutations α_e, α_r are given by $\alpha_e(e) = e$, $\alpha_e(r) = r$ and $\alpha_r(e) = r$, $\alpha_r(r) = e$. Thus the left regular representation is $D : C_2 \rightarrow GL(2, \mathbb{C})$ with $D(g) = P_g$ for:

$$P_e = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad P_r = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Similarly, let $C_3 = \{e, r, r^2\}$. By a similar calculation, we can see that the left regular representation is:

$$P_e = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad P_r = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \quad P_{r^2} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}.$$

7. Suppose that $D : G \rightarrow GL(n, \mathbb{C})$ is a representation. Show that $\bar{D} : G/\text{Ker}(f) \rightarrow GL(n, \mathbb{C})$ given by $\bar{D}(g\text{Ker}(f)) = D(g)$ is a faithful representation of $G/\text{Ker}(f)$.

Solution: First, we need to check that $\bar{D}$ is well-defined. Let $g\text{Ker}(f) = g'\text{Ker}(f)$. Then $g' = gk$ for some $k \in \text{Ker}(f)$, which gives:

$$\bar{D}(g'\text{Ker}(f)) = D(g') = D(gk) = D(g)D(k) = D(g) = \bar{D}(g\text{Ker}(f)),$$

thus the map is well-defined. Further, we have that it is a homomorphism:

$$\bar{D}(g\text{Ker}(f)g'\text{Ker}(f)) = \bar{D}((gg')\text{Ker}(f)) = D(gg') = D(g)D(g') = \bar{D}(g\text{Ker}(f))\bar{D}(g'\text{Ker}(f)).$$

Equivalence and inequivalence of linear representations

8. Show that the following maps extend to representations of $C_3 = \{e, r, r^2\}$:

$$D_1(r) = \begin{pmatrix} e^{2\pi i/3} & 0 \\ 0 & e^{4\pi i/3} \end{pmatrix}, \quad D_2(r) = \begin{pmatrix} e^{4\pi i/3} & 0 \\ 0 & e^{2\pi i/3} \end{pmatrix}.$$

Show that they are equivalent by explicitly identifying an intertwining matrix.

Solution: Evidently, both satisfy $D_1(r)^3 = I$, $D_2(r)^3 = I$. This ensures that both of these choices extend to representations of C_3 . The intertwining matrix we can use is:

$$S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = S^{-1}.$$

Checking this for the generating matrices, we have:

$$\begin{aligned} SD_1(r)S &= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} e^{2\pi i/3} & 0 \\ 0 & e^{4\pi i/3} \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & e^{2\pi i/3} \\ e^{4\pi i/3} & 0 \end{pmatrix} = \begin{pmatrix} e^{4\pi i/3} & 0 \\ 0 & e^{2\pi i/3} \end{pmatrix} \\ &= D_2(r). \end{aligned}$$

9. Show that the following maps extend to representations of $C_4 = \{e, r, r^2, r^3\}$:

$$D_1(r) = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \quad D_2(r) = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

Show that the representations are equivalent, and that the general intertwining matrix between them is given by:

$$S = \begin{pmatrix} a & b \\ ia & -ib \end{pmatrix},$$

where a, b are non-zero complex numbers.

Solution: We obviously have that $D_1(r)^4 = I$, since the matrix is diagonal and $i^4 = 1, (-i)^4 = 1$. For the second matrix, we have:

$$D_2(r)^2 = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix},$$

which obviously squares to I . To find the intertwining matrix S , we let:

$$S = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

We demand that for the generators we have:

$$D_1(r) = S^{-1}D_2(r)S.$$

But, it is much easier to check that:

$$SD_1(r) = D_2(r)S.$$

Inserting the above formulae, we have:

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} \quad \Rightarrow \quad \begin{pmatrix} ia & -ib \\ ic & -id \end{pmatrix} = \begin{pmatrix} c & d \\ -a & -b \end{pmatrix}.$$

Comparing the entries, we see that $c = ia$ and $d = -ib$. This guarantees the top rows and bottom rows are equal in the above matrices. Thus, we get the general form of S . The only further requirement is the $\det(S) \neq 0$, so that the matrix is invertible. This gives $0 \neq a(-ib) - b(ia) = -2iab$, requiring that a, b are both non-zero.

10. Prove that for any 2×2 complex matrices A, B , we have $\text{Tr}(AB) = \text{Tr}(BA)$. Can you extend your proof to $n \times n$ complex matrices? [Hint: if $C = AB$ is the product of two matrices A, B , how can we express C_{ij} (the (i, j) th entry of C) in terms of the entries of A, B ?]

Solution: We do it in the general $n \times n$ case. We have:

$$\begin{aligned}\text{Tr}(AB) &= \sum_{i=1}^n \sum_{j=1}^n A_{ij} B_{ji} \\ &= \sum_{i=1}^n \sum_{j=1}^n B_{ji} A_{ij} \\ &= \sum_{j=1}^n \sum_{i=1}^n B_{ji} A_{ij} \\ &= \text{Tr}(BA).\end{aligned}$$

11. For the maps in Question 4 which are representations, evaluate their characters χ_D in each case.

Solution: The first three maps are representations:

- (a) We have $\chi_D(n) = 2^n + 3^n$.
- (b) We have $\chi_D(n) = 2$ for all n . In particular, we see that the representations in (a) and (b) are inequivalent representations of $\mathbb{Z}$.
- (c) This time, we need to calculate all the matrices. We certainly have:

$$D(0) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad D(1) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad D(2) = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}, \quad D(3) = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

Hence, the character is $\chi_D(0) = 2, \chi_D(1) = 0, \chi_D(2) = -2$ and $\chi_D(3) = 0$.

12. Show that the following map extends to a representation of $C_4 = \{e, r, r^2, r^3\}$:

$$D_3(r) = \begin{pmatrix} i & 0 \\ 0 & i \end{pmatrix}.$$

Is it equivalent to the representations in Question 9?

Solution: It obviously extends to a representation of C_4 , since $D_3(r)^4 = I$ is evident from the diagonal form of $D_3(r)$. It is inequivalent to the representations from Question 9: we can check the characters. For the matrices in Question 9, we have $\chi_{D_1}(r) = \chi_{D_2}(r) = 0$. However, for this matrix, we have $\chi_{D_3}(r) = 2i$.

13. Let $D : G \rightarrow GL(n, \mathbb{C})$ be a representation of the group G . Prove that:

- (a) $\chi_D(e) = n$;
- (b) if x, y lie in the same conjugacy class in G , then $\chi_D(x) = \chi_D(y)$.

Solution:

- (a) We always have $D(e) = I$ since D is a homomorphism, so $\chi_D(e) = \text{Tr}(I) = n$.
- (b) Since x, y lie in the same conjugacy class, we have $y = gxg^{-1}$ for some $g \in G$. Thus we have:

$$\chi_D(y) = \chi_D(gxg^{-1}) = \text{Tr}(D(gxg^{-1})) = \text{Tr}(D(g)D(x)D(g)^{-1}) = \text{Tr}(D(x)) = \chi_D(x).$$